

Project Overview

This project, developed as a final assignment for the **UPC Postgraduate Program in Deep Learning**, aims to build an **AI-powered semantic analysis system for construction site visit reports** (“actas de obra”). These reports are typically written in **Microsoft Word format**, contain **technical and domain-specific language**, and are largely **unstructured**.

The core objective is to design and implement a **realistic, end-to-end NLP pipeline** that transforms raw technical documents into **searchable knowledge**, and enables **context-aware question answering** using modern Transformer-based models.

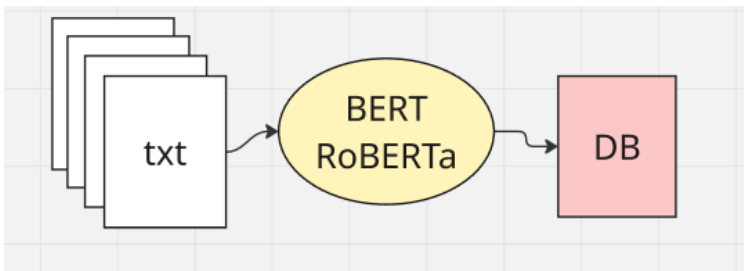
The system follows a **Retrieval-Augmented Generation (RAG)** paradigm, grounded in Transformer Encoder–Decoder principles.

Conceptual Foundation: Transformer Architecture

At the theoretical level, the project is grounded in the **Transformer architecture**, which is composed of two main blocks:

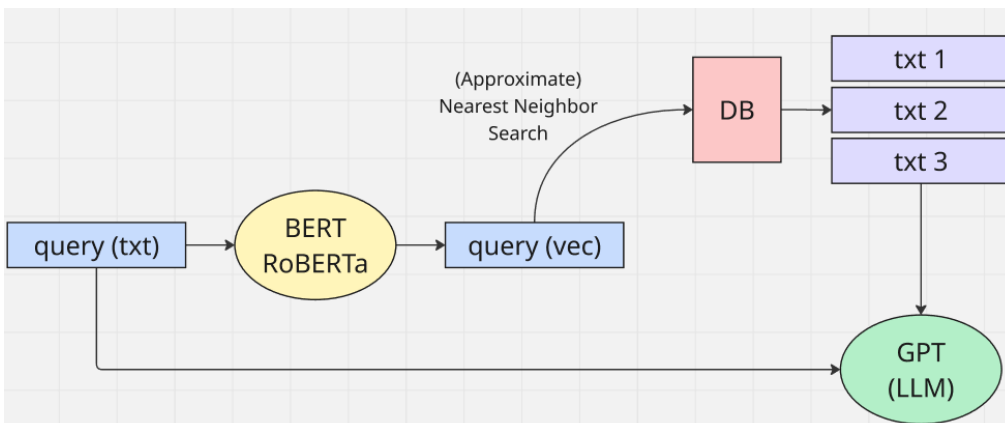
Encoder

- The **Encoder** is responsible for **reading and understanding text**.
- It converts input text into dense vector representations (embeddings) that capture **semantic meaning**.
- In this project, **BERT or RoBERTa** models (pre-trained Transformers) play the role of the **Encoder**.



Decoder

- The **Decoder** is responsible for **generating language output** based on:
 - The user query.
 - The encoded information retrieved from the database.
- In this architecture, a **GPT-style Large Language Model (LLM)** acts as the **Decoder**, producing fluent, contextualized answers.



Step-by-Step System Architecture

1. Document Ingestion and Encoding (Top Left of Diagram)

- A collection of **text documents** (initially public datasets, later Word-based construction site visit reports) is processed.
- Each document is passed through **BERT or RoBERTa**.
- The model outputs a **vector embedding** for each document or document chunk.

Key point for the presentation:

BERT/RoBERTa acts purely as an **Encoder** that learns semantic representations of technical text.

2. Vector Database: Keys and Values

The embeddings are stored in a **vector database** structured conceptually as:

- **Key**: embedding vector (semantic representation)
- **Value**: original text chunk.

This structure allows:

- Fast retrieval
- Traceability back to the original report
- Explainability of results

3. Approximate Nearest Neighbor (ANN) Search

Because embeddings are high-dimensional, search is inefficient.

The project uses **Approximate Nearest Neighbor (ANN) search** to:

- Efficiently retrieve the most semantically similar text chunks
- Scale to large document collections

When a query is issued:

1. The database performs ANN search
2. The **top-k most relevant text segments** are returned

This step is critical to ensure **performance and scalability**, and is an important Deep Learning systems topic.

4. Query Processing Pipeline

- A **user query written in natural language** enters the system.
- The query is passed through **the same BERT/RoBERTa Encoder**.
- This ensures the query embedding lives in the **same vector space** as the documents.

This symmetry between document embeddings and query embeddings is fundamental to semantic search.

5. Retrieval-Augmented Generation (RAG)

The retrieved text chunks (txt1, txt2, txt3 in the diagram) are then:

- Injected as **context**
- Along with the original user query
- Into a **GPT-style LLM (Decoder)**

The LLM:

- Generates answers grounded **only in retrieved documents**
- Produces structured, coherent, and fluent responses

Role of Pretrained BERT / RoBERTa

The project explicitly leverages **transfer learning**:

- BERT or RoBERTa is used **pretrained** on large public corpora.
- Initially, embeddings are generated without task-specific tuning.
- This enables fast experimentation and strong semantic baselines.

Training Strategy: From Scratch + Fine-Tuning

The project includes **two learning phases**, which is academically important:

1. Transformer Defined from Scratch (Educational Component)

- A **Transformer architecture will be implemented from scratch** using:
 - Public datasets
- This step demonstrates:
 - Understanding of self-attention
 - Encoder–Decoder mechanics
 - Positional encoding and layers

2. Fine-Tuning with Real Construction Reports

- The system is then **fine-tuned** using:
 - Word documents containing **real construction site visit reports**
- Domain-specific vocabulary (electrical installations, safety issues, compliance) is learned.
- The embeddings become specialized for **construction and engineering language**.

This bridges theory with real-world industrial data.

